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Plant Disease Detection Using an Innovative Swin-Axial Transformer

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ABSTRACT Plant diseases are a significant threat to global agricultural production, and accurate and efficient disease detection is crucial for ensuring food security. With the rapid development of deep learning technology, Transformer architectures have demonstrated strong global feature extraction capabilities in image classification and detection tasks. However, traditional Transformer models still face high computational costs and large parameter volumes when handling multi-scale diseases and complex backgrounds. To address this challenge, this paper proposes the Swin-Axial Transformer model based on the Swin Transformer. By introducing the TokenEmbedder module, the number of tokens is reduced, and multi-scale deep convolution is used to efficiently extract image features, significantly lowering computational costs. Furthermore, the Axial Transformer module reduces computational complexity by restricting self-attention calculations to local regions along the axial direction. With the combination of the Axial Compression module and the Detail Enhancement module, the model can efficiently extract global semantic features and effectively supplement local detail information. Compared to traditional global self-attention mechanisms, the Axial Transformer successfully avoids computational bottlenecks through axial compression strategies. Experimental results show that the model achieves a precision of 79.52% on the PlantDoc dataset, 99.79% on the PlantVillage dataset, and 95.69% on a self-constructed fusion dataset. The model's parameter size is 14.87 M with a computational load of 2.96 GMac, which represents a reduction of 46% and 32%, respectively, compared to the Swin Transformer-T model, significantly improving computational efficiency. The model presented in this paper provides an efficient and reliable solution for large-scale crop disease detection.

INDEX TERMS Plant disease detection, deep learning, global feature extraction, multi-scale features.

I. INTRODUCTION

Plant diseases pose a significant challenge to modern agriculture, with globally estimated grain yield reduction caused by diseases exceeding 10%. Timely and effective disease detection is crucial for preventing the spread of diseases and mitigating their impacts. However, plant disease detection presents unique challenges compared to other object or pattern detection tasks. Disease symptoms often vary significantly across different plant species, and their manifestations are strongly influenced by environmental factors such as changes in light, temperature, and humidity. This makes plant disease detection more complex and challenging. Currently, most disease detection still relies on manual field evaluations by crop pathology experts, a process that is not only subjective, time-consuming, and labor-intensive but also prone to omissions and misjudgments. By contrast, machine learning, particularly deep learning methods, has emerged as a vital tool for plant disease diagnosis. In recent years, image classification techniques have been widely applied in crop disease detection. Traditional image classification methods often depend on manually designed features, whose selection and extraction accuracy directly affect classification performance. Factors such as lighting variations and background noise in natural environments further degrade image quality, exacerbating the impact on classification results.

In recent years, machine learning methods, particularly those used for image classification, have emerged as vital tools for crop disease diagnosis. Common techniques include image segmentation, feature extraction, and pattern classification to categorize disease images. For

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example, Kamal et al. [1] employed support vector machines (SVM) to classify palm oil leaf disease images, achieving an accuracy of 97%. Phadikar et al. [2] compared Bayesian classifiers with SVM for classifying rice leaf diseases, finding that the Bayesian classifier outperformed SVM in both recognition accuracy and processing speed. Furthermore, Panchal et al. [3] combined K-means clustering with HSV-dependent classification methods to improve the accuracy of leaf infection area recognition. Tabbakh et al. [4] introduced a machine learning approach that combines wavelet transforms, GLCM, and statistical features for plant disease classification. By applying the SMOTE method to address data imbalance and using the LGBM model, they achieved a classification accuracy of 94.39%. Bhargava et al. [5] employed logistic regression (LR) and SVM methods for classifying vegetable and fruit leaf varieties, improving accuracy via PCA feature selection. Their study also enhanced image quality through Gaussian filtering and utilized fuzzy C-means clustering and grab-cut methods for image segmentation, effectively extracting leaf features. However, traditional machine learning methods depend on manual feature extraction, and their performance is heavily influenced by the accuracy of feature selection. Additionally, manual feature design is subjective and struggles to manage the complexity of varying disease features. Disease images taken in natural environments are also often affected by factors such as uneven lighting, which degrades image quality and impacts classification results.

In contrast to traditional machine learning methods, deep learning approaches have demonstrated stronger feature extraction capabilities in plant disease classification. Convolutional neural networks (CNNs) are widely regarded as one of the most effective methods for disease identification. CNNs can automatically extract relevant features from training data without the need for manual feature design, greatly simplifying the disease detection process. Falaschetti et al. [6] proposed a CNN-based classification model that enabled real-time classification on resource-constrained platforms. Vishnoi et al. [7] enhanced classification accuracy for apple leaf diseases through data augmentation and a lightweight design. Khattak et al. [8] introduced a deep learning model based on CNN for citrus fruit and leaf disease detection, achieving a classification accuracy of 94.55%. Roy et al. [9] proposed a crop growth stage detection model that fuses DenseNet with YOLOv4, demonstrating high precision and efficiency in complex environments. Liang et al. [10] designed a plant disease diagnosis and severity estimation network (PD^2SE-Net) based on residual networks and Shuffle-Net, which improved disease detection accuracy by enhancing feature extraction capabilities. Johnson et al. [11] applied Mask R-CNN and residual networks to identify blight spots on potato leaves, effectively distinguishing infected areas from the background. Additionally, this study created independent models for each color space by converting the dataset into multiple color spaces. Hassan et al. [12] introduced a deep learning model based on Inception layers and residual connections, employing depthwise separable convolutions to reduce the number of parameters and lower computational costs. Due to its efficient feature extraction capabilities, CNN has become the dominant technology in plant disease classification.

Recently, the Vision Transformer (ViT) model, inspired by the Transformer architecture, has achieved significant success in image classification [13]. Tabbakh et al. [14] introduced the TLMViT model, which combines transfer learning with ViT, enhancing plant disease classification through multi-stage feature extraction and data augmentation, resulting in validation accuracies of 98.81% and 99.86%. Gole et al. [15] designed the lightweight ViT network TrIncNet, replacing the computationally expensive MLP module in ViT, significantly improving both computational efficiency and classification performance for plant disease detection tasks. Chen et al. [16] proposed the FasterViT model, which combines CNN's local learning with ViT's global information processing ability, achieving excellent performance on small sample plant disease datasets while reducing training time. Zhang et al. [17] applied the Swin-Transformer to rice disease diagnosis, achieving an accuracy of 93.4%. Roy et al. [18] developed the DenseSPH-YOLOv5 model, which integrates DenseNet and Swin-Transformer prediction heads, significantly improving crop disease detection, particularly in multi-scale and complex backgrounds. Thai et al. [19] proposed a ViT-based crop leaf disease detection model, applied to cassava leaf disease classification. Experimental results showed that this model improved accuracy by at least 1% compared to traditional CNN models such as EfficientNet and ResNet50d on the cassava leaf disease dataset. Borhani et al. [20] proposed a combined ViT and traditional CNN method, tested and evaluated for plant disease classification tasks, finding that while ViT improved accuracy, it reduced prediction speed. The approach effectively compensated for this speed reduction by combining attention blocks with CNN modules.

Despite the significant progress made by Transformer-based models in plant disease detection, existing methods still face challenges such as high computational costs and model complexity, especially when handling complex backgrounds and multi-scale disease images, where traditional models tend to encounter issues with excessive computational resource consumption. To address these challenges, this paper proposes the Swin-Axial Transformer model based on the Swin Transformer architecture. The core innovation lies in combining the TokenEmbedder module and Axial Transformer,

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successfully overcoming the shortcomings of existing methods in terms of computational efficiency, feature extraction, and multi-scale modeling. The details are as follows:

1) Proposing the TokenEmbedder module: Traditional Transformer models experience a significant increase in computational complexity and memory consumption due to the large number of input tokens when handling high-resolution images. To solve this problem, this paper proposes the TokenEmbedder module. This module reduces the number of tokens and uses multi-scale deep convolutions to efficiently extract image features, thereby significantly reducing computational complexity and memory consumption. TokenEmbedder converts the input features into a fixed number of latent tokens, retaining key feature information while improving feature extraction efficiency, making it particularly suitable for high-resolution image processing. By reducing the number of tokens and using efficient convolution operations, the TokenEmbedder module provides more efficient input for subsequent Transformer modules, thereby improving overall computational efficiency.
2) Introduction of the Axial Transformer Module: Traditional Transformer models use a global self-attention mechanism to capture long-range dependencies. However, when processing large-scale images, this approach significantly increases computational complexity, thus limiting its application scope. To address this issue, this paper introduces the Axial Transformer module. This module reduces computational complexity by restricting self-attention calculations to local regions along the axial directions. By combining the Axial Compression module and the Detail Enhancement module, it not only efficiently extracts global semantic features but also effectively supplements local detail information. The synergy of these two modules significantly enhances the feature modeling capabilities of the Axial Transformer module in complex backgrounds and multi-scale scenarios.
3) Performance improvement: Compared with the existing state-of-the-art models, the Swin-Axial Transformer shows significant improvements in several quantitative metrics. For example, the Swin-Axial Transformer achieves 79.52% precision on the PlantDoc dataset, improving 5.08% over the T-CNN (ResNet-101) model's 74.44%; on the PlantVillage dataset, the model achieves 99.79% precision, 1.07% better than the TLMViT model's 98.72%; and on the self-constructed fusion dataset, the model achieves 95.69% precision, 2.45% better than the Swin Transformer-T model's 93.34%. The model's parameter size is 14.87 M, and the computation is 2.96 GMac, reducing by 46% and 32%, respectively, compared to the Swin Transformer-T model, significantly improving computational efficiency. These experimental results show that the Swin-Axial Transformer not only surpasses several mainstream models in terms of accuracy but also shows a significant advantage in computational resource consumption.

II. Swin-Axial Transformer Model

The Swin-Axial Transformer model integrates the advantages of the Swin Transformer, TokenEmbedder, and Axial Transformer to create an efficient deep learning architecture. As depicted in Figure 1, the architecture consists of four stages. The first three stages for feature processing are inspired by the design philosophy of the Swin Transformer, incorporating the Shifted Window Attention mechanism to efficiently extract image features. By progressively reducing the image size and increasing the number of feature channels, hierarchical feature representation is achieved, enabling the extraction of rich semantic information at multiple levels.

In Stage 1, the input image ($H \times W \times 3$) is first divided into several non-overlapping 4×4 patches. Each patch is transformed into a high-dimensional feature representation via a Linear Embedding operation. These features are then passed through the Swin Transformer Block, where the Shifted Window Attention mechanism is applied for self-attention computation, effectively capturing local information. The processed output feature has the size $\frac{\mathrm{H}}{4} \times \frac{\mathrm{W}}{4} \times 48$, with 48 representing the embedding dimension.

In Stage 2, a Patch Merging operation is performed, gradually reducing the size of the feature map while increasing the number of channels. This operation not only effectively reduces the computation load but also enhances the feature representation capabilities. After this operation, the feature map size becomes $\frac{\mathrm{H}}{8} \times \frac{\mathrm{W}}{8} \times 2\mathrm{C}$, where C is the number of channels. Next, the features pass through another Swin Transformer Block and continue self-attention calculations with the Shifted Window Attention mechanism, further extracting deeper local features.

In Stage 3, the feature map processed in the previous stages undergoes another Patch Merging operation, resulting in an output feature map of size $\frac{H}{16} \times \frac{W}{16} \times 4C$. At this stage, the feature map dimensions further increase, strengthening the model's feature representation. Then, the features are processed through another Swin Transformer Block with multiple Shifted Window Attention operations (six times in this stage), extracting global information and enhancing the model's perception of global features.

In Stage 4, the features are passed to the TokenEmbedder module we designed. This module uses multi-scale deep convolution operations to further compress and extract features, converting them into token

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representations suitable for subsequent classification tasks. By reducing the number of tokens, the TokenEmbedder effectively reduces the model's computational complexity while preserving its feature representation capacity. This module transforms the feature representations (*B, C, H, W*), where *B* is the batch size, *C* is the number of channels, and *H* and *W* are the height and width of the feature map, into a fixed-size token representation. The output size is (*B, T, E*), where *T* is the number of tokens generated, and E is the embedding dimension of each token.

Finally, in the fourth stage, we introduce the Axial Transformer module. By combining Axial-Squeeze Attention, it achieves efficient fusion of local and global features, further improving the model's feature extraction capacity and classification performance. This design not only effectively captures global semantic information but also retains key local features, enhancing the model's adaptability and flexibility in various scenarios.

The Swin-Axial Transformer model, by combining the Swin Transformer Block, TokenEmbedder module, and Axial Transformer module, significantly enhances the model's performance and computational efficiency. The Swin Transformer Block integrates a local self-attention mechanism with a window shifting strategy, effectively avoiding the high computational overhead associated with global self-attention mechanisms. Compared to traditional Transformer models with a computational complexity of $O(N^2)$ (where N is the number of tokens), Swin Transformer reduces the complexity to $O(N \times M)$, where M is the window size, making the model highly efficient for processing large-scale high-resolution images. The TokenEmbedder module enhances computational efficiency by reducing the number of tokens and using multi-scale deep convolution. It reduces computational complexity and memory consumption by converting input features into a fixed number of latent tokens while using deep convolution to effectively improve feature extraction efficiency, especially for high-resolution image processing. The Axial Transformer module further boosts computational efficiency by introducing the Axial-Squeeze Attention mechanism. Unlike traditional global self-attention mechanisms, the Axial Transformer limits attention computation to local regions along the axial directions (horizontal and vertical), significantly reducing computational complexity. This avoids the $O(N^2)$ computational cost and reduces it to $O(N)$, enabling the model to run efficiently on larger datasets and higher-resolution images.

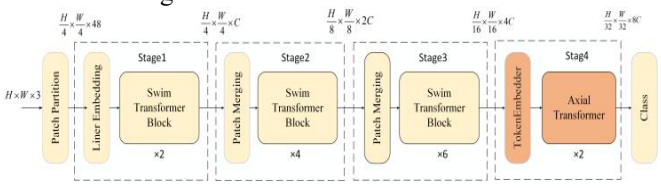


FIGURE 1. **Architecture diagram of the Swin-Axial Transformer model**

A. TokenEmbedder Module

In plant disease detection tasks, the complexity and diversity of disease features impose higher requirements on feature extraction. Existing Transformer models typically divide images into fixed-size patches and transform them into tokens for processing. While this method is effective in capturing global features, it also introduces a significant amount of redundant computation, resulting in increased computational complexity and memory consumption. Additionally, traditional methods have limited performance when handling multi-scale and local features, making it difficult to meet the practical demands of plant disease detection tasks.

In order to solve the issue, this paper proposes the TokenEmbedder module (Figure 2). It improves feature extraction efficiency and reduces the number of tokens to ensure that the Transformer model can process high-dimensional input features efficiently. Specifically, the TokenEmbedder module converts input features $X \in \mathbb{R}^{H \times W \times C}$ into a fixed number of latent tokens $T \in \mathbb{R}^{N \times E}$,where H, W, C denote the height, width, and channel number of the input features, N represents the number of tokens, and E is the embedding dimension.

First, the module performs an initial processing of the input features X through a 3 × 3 convolution, extracting a feature map that captures basic spatial information and adjusting the channel number to match the subsequent embedding dimension, resulting in the preliminary feature F_{init}. This initial convolution operation is expressed by the following formula:

$$F_{\text{init}} = \text{Conv}_{3\times3}(X),\ F_{\text{init}} \in \mathbb{R}^{B \times C \times H \times W} \tag{1}$$

Where B, H, W represent batch size, height, and width, respectively.

On this basis, to capture multi-scale features, depthwise convolutions with different receptive fields are designed, including kernels of size 5 × 5, 7 × 7, and 9 × 9. An "Identity" path (identity mapping) is also retained to preserve the original features. The output feature F_{k} is expressed by the following formula:

$$F_{\text{k}} = \text{DWConv}_{\text{k}\times\text{k}}(F_{\text{init}}),\ k \in \{5,7,9\} \tag{2}$$

Subsequently, all output features are fused through channel concatenation, using a 1 × 1 convolution to complete the fusion process, resulting in the multi-scale feature F_{multi}, which is expressed as:

$$F_{\text{multi}} = \text{Conv}_{1\times1}\left(\text{Concat}\left(F_5, F_7, F_9, F_{\text{identity}}\right)\right),\quad F_{\text{multi}} \in \mathbb{R}^{B \times C \times H \times W} \tag{3}$$

This design adopts a multi-scale convolution kernel structure without dilation. Compared to dilated or large kernel convolutions, it not only maintains sufficient receptive fields but also avoids background noise and feature sparsity issues. Additionally, by incorporating the "Identity" path, this module further enhances the consistency and balance of feature representations,

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providing strong support for multi-scale feature modeling in complex scenarios.

After obtaining the multi-scale feature F_{multi}, the module introduces an attention mechanism to further highlight the key features. First, a 1 × 11 depthwise convolution is used to compute the feature weights in the channel direction, and then a 11 × 1 convolution is applied to capture the spatial relationships of the features. Next, a 1 × 1 convolution is used for channel integration. Finally, the attention weights are mapped to the range of [0, 1] using the Sigmoid activation function, generating the final attention weights A, The expression is as follows:

$$A = \sigma\Big(\text{DWConv}_{1\times11}\big(\text{Conv } 1\times 1\big(\text{DWConv}_{11\times1}(F_{multi})\big)\big)\Big),\ A \in \mathbb{R}^{B\times C_{\text{multi}}\times H\times W} \quad (4)$$

Subsequently, the attention weight A is multiplied with the multi-scale feature F_{multi} to obtain the refined feature F_{att}, The expression is as follows:

$$F_{\text{att}} = A \odot F_{\text{multi}} \quad (5)$$

Finally, the module passes the enhanced feature through a linear transformation layer to map it to a fixed number of latent tokens, The expression is as follows:

$$F_{\text{T}} = \text{Linear}\,(F_{\text{att}}),\ T \in \mathbb{R}^{B\times T\times E} \quad (6)$$

Here, T represents the number of tokens, and E represents the embedding dimension of each token. Through this process, the TokenEmbedder module achieves significant reductions in computational complexity while retaining essential multi-scale and global features. This provides more efficient input representations for the subsequent Transformer module, enabling improved overall processing.

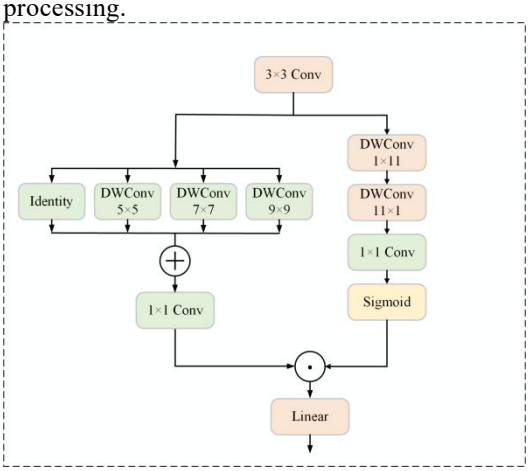


FIGURE 2. **TokenEmbedder module structure diagram**

B. Axial Transformer Module

In complex tasks such as plant disease detection, models must be capable of performing both local and global feature modeling to enhance performance. However, traditional global self-attention mechanisms (e.g., Transformers), while effective in capturing long-range dependencies, exhibit a computational complexity of $O(N^2)$, which grows quadratically with increasing input resolution. This significantly limits the applicability of traditional methods in high-resolution image processing tasks.

To address this challenge, this paper introduces the Axial-Squeeze Attention module (shown in Figure 4), which combines an Axial Compression module with a Detail Enhancement module. The module aims to efficiently extract global semantic features while supplementing local detail information. The synergistic interaction between these two components greatly improves the network's feature modeling capabilities in complex backgrounds and multi-scale scenarios.

Building upon this, the paper introduces the Axial Transformer (shown in Figure 3), which integrates a standard feedforward network (MLP) and layer normalization with thexial-Squeeze Attention mechanism. This integratio An enables the effective modeling of diverse features. While preserving efficient feature extraction capabilities, this approach significantly reduces computational complexity, providing an effective solution for feature representation in large-scale tasks.

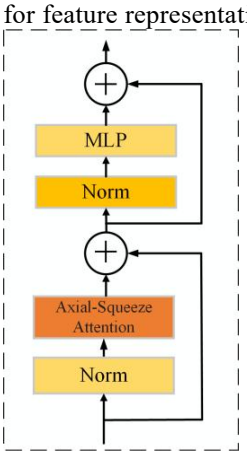


FIGURE 3. **Axial Transformer module structure diagram**

1) AXIAL-SQUEEZE ATTENTION

a: Axial Compression Module

First, the input feature $X \in \mathbb{R}^{C\times H\times W}$ generates the Query (Q), Key (K), and Value (V) matrices through three linear layers, expressed by the formulas:

$$Q \in \mathbb{R}^{C_{qk}\times H\times W} = W_{\text{q}}X \quad (7)$$

$$K \in \mathbb{R}^{C_{qk}\times H\times W} = W_{\text{k}}X \quad (8)$$

$$V \in \mathbb{R}^{C_{v}\times H\times W} = W_{\text{v}}X \quad (9)$$

Where W_{q}, W_{k}, W_{v}, are learnable parameters. Next, in the horizontal direction (i.e., the W-direction), horizontal compression is achieved by extracting the average values of the Q, K, V feature maps:

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$$Q_{(h)} \in \mathbb{R}^{C_{qk} \times H} = \frac{1}{W}\left(Q \in \mathbb{R}^{C_{qk} \times H \times W} \cdot I_W\right) \quad (10)$$

$$K_{(h)} \in \mathbb{R}^{C_{qk} \times H} = \frac{1}{W}\left(K \in \mathbb{R}^{C_{qk} \times H \times W} \cdot I_W\right) \quad (11)$$

$$V_{(h)} \in \mathbb{R}^{C_{v} \times H} = \frac{1}{W}\left(V \in \mathbb{R}^{C_{v} \times H \times W} \cdot I_W\right) \quad (12)$$

Here, $I_W \in \mathbb{R}^{W \times 1}$ is a vector of all 1s with length W. The matrix multiplication sums the elements in the W-direction and then divides by $\frac{1}{W}$ to obtain the average value of all features in the W-direction.

In the vertical direction (i.e., the H-direction), vertical compression is achieved by extracting the average values of the Q, K, and V feature maps:

$$Q_{(v)} \in \mathbb{R}^{C_{qk} \times W} = \frac{1}{H}\left(Q \in \mathbb{R}^{C_{qk} \times W \times H} \cdot I_H\right) \quad (13)$$

$$K_{(v)} \in \mathbb{R}^{C_{qk} \times W} = \frac{1}{H}\left(K \in \mathbb{R}^{C_{qk} \times W \times H} \cdot I_H\right) \quad (14)$$

$$V_{(v)} \in \mathbb{R}^{C_{v} \times W} = \frac{1}{H}\left(V \in \mathbb{R}^{C_{v} \times W \times H} \cdot I_H\right) \quad (15)$$

Where, $I_H \in \mathbb{R}^{H \times 1}$ is a vector of all 1s with length H. The matrix multiplication sums the elements in the H-direction and then divides by $\frac{1}{H}$ to obtain the average value of all features in the H-direction.

The compressed Query, Key, and Value matrices model the long-term dependencies of the corresponding axes through the self-attention mechanism in the horizontal and vertical directions, respectively:

$$Y_{(h)} = \text{softmax}\left(\frac{Q_{(h)}K_{(h)}^{T}}{\sqrt{d_k}}\right)V_{(h)} \quad (16)$$

$$Y_{(v)} = \text{softmax}\left(\frac{Q_{(v)}K_{(v)}^{T}}{\sqrt{d_k}}\right)V_{(v)} \quad (17)$$

Where d_k is the scaling factor of the feature dimension, used to balance the attention distribution.

Finally, the enhanced features in the horizontal and vertical directions are integrated through a fusion operation, and then a 3 × 3 depth-wise convolution layer further compresses the channel information to generate the final output feature Y_{global}:

$$Y_{global} = \text{DWConv}_{3\times3}\left(Y_{(h)} + Y_{(v)}\right) \quad (18)$$

This design reduces the computational complexity of the traditional self-attention mechanism from $O(N^2)$ to $O(N)$, while still retaining the ability to capture global dependency relationships.

b: Detail Enhancement Module

While the axial compression operation obtains global features, it may ignore some local details. To solve this problem, a detail-enhancement module is designed to enhance spatial detail features. As shown in Figure 4, first, a set of additional parameters $W_q^{(e)}$, $W_k^{(e)}$, and $W_v^{(e)}$ are applied to the input $X \in \mathbb{R}^{C \times H \times W}$ to obtain the corresponding $Q_{(e)}$, $K_{(e)}$, $V_{(e)}$:

$$Q_{(e)} \in \mathbb{R}^{C_{qk} \times H \times W} = W_q^{(e)}X \quad (19)$$

$$K_{(e)} \in \mathbb{R}^{C_{qk} \times H \times W} = W_k^{(e)}X \quad (20)$$

$$V_{(e)} \in \mathbb{R}^{C_{v} \times H \times W} = W_v^{(e)}X \quad (21)$$

Then, the three are concatenated in the channel dimension and passed to 1 × 3 and 3 × 1 depthwise-convolution modules to complete the detail enhancement:

$$X_{qkv} \in \mathbb{R}^{(2C_{qk}+C_v) \times H \times W} = \text{Concat}\left(Q_{(e)} + K_{(e)} + V_{(e)}\right) \quad (21)$$

$$X_{qkv} \in \mathbb{R}^{(2C_{qk}+C_v) \times H \times W} = \text{DWConv}_{3\times1}\left(\text{DWConv}_{1\times3}\left(X_{qkv}\right)\right) \quad (22)$$

Among them, the 1 × 3 depthwise-convolution is used to capture spatial details in the width direction, and the 3 × 1 depthwise-convolution is used to capture spatial details in the height direction. In this way, more fine-grained local information can be extracted from the auxiliary Q, K, V features.

Then this module compresses the features from $2C_{qk} + C_v$ dimensions to C dimensions through a 1 × 1 convolution and generates the detail-enhanced feature $Y_{enhanced}$:

$$Y_{enhanced} \in \mathbb{R}^{C \times H \times W} = \text{Conv}_{1\times1}\left(X_{qkv}\right) \quad (23)$$

Finally, the output Y_{global} of the axial-compression module generates a weight through the Sigmoid function, and then performs a point-wise multiplication with the detail-enhanced feature $Y_{enhanced}$ to obtain the fused output. The calculation formula is:

$$output = \text{sigmoid}\left(Y_{global}\right) \odot Y_{enhanced} \quad (24)$$

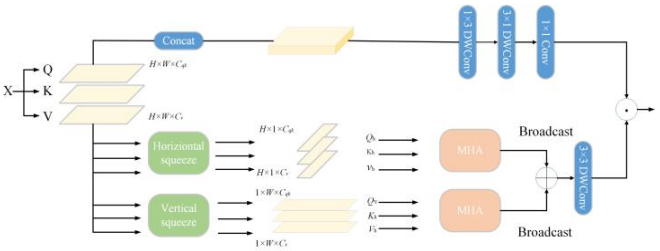


FIGURE 4. Axial-Squeeze attention model architecture

III. Experiments

A. Experimental Datasets

This study selected three datasets to validate the effectiveness and robustness of the proposed model. First, the PlantDoc dataset was used as the primary experimental dataset to assess the model's performance in complex scenarios [21]. Secondly, the PlantVillage dataset and a self-constructed fusion dataset were chosen to further verify the model's feature extraction capability and generalization ability in standardized environments. By testing on these three datasets,

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we comprehensively analyzed the model's effectiveness and adaptability in real-world applications.

1) PLANTDOC DATASET

The PlantDoc dataset is designed for plant disease detection and is based on real-world field scenarios. It comprises 13 major crops and 27 disease types, totaling 2,598 high-resolution, clearly annotated images. Collected from natural environments, the dataset includes complex backgrounds, lighting variations, and occluded leaves, making it highly representative of real-world application scenarios. Due to its diverse disease types and unique environmental conditions, the PlantDoc dataset is widely used to evaluate the performance of deep learning models in plant disease classification and detection tasks. It serves as a valuable resource for agricultural disease diagnosis research. Figure 5 presents some sample images from the PlantDoc dataset.



FIGURE 5. Image examples from the PlantVillage dataset

2) PLANTVILLAGE DATASET

The PlantVillage dataset is a publicly available resource specifically designed for plant disease detection and classification tasks. It consists of 14 crop types and 38 disease types, totaling 54,306 high-quality, clearly annotated leaf images. The dataset was collected under controlled laboratory conditions, featuring a uniform background with minimal noise, which makes it well-suited for studies involving supervised learning and transfer learning. However, due to its idealized environment, the dataset's generalization to real-world scenarios is limited. It is widely used for evaluating the performance of models in plant disease classification and serves as one of the key benchmark datasets in plant disease research. Figure 6 presents some representative image samples from the PlantVillage dataset.



FIGURE 6. Image examples from the PlantVillage dataset

3) SELF-CONSTRUCTED FUSION DATASET

The self-constructed fusion dataset is created by integrating public datasets of healthy and diseased samples from three crops: lemon, rice, and wheat. It encompasses 16 disease types, with images sourced from various origins involving different plant species. The dataset accounts for factors such as lighting variations, background changes, and seasonal transitions. This approach aims to enhance the model's generalization ability, improving its adaptability to a wide range of scenarios. Figure 7 presents some representative image samples from the self-constructed fusion dataset.



FIGURE 7. Image examples from the Self-Constructed Fusion dataset

B. Experimental Environment and Parameter Settings

The experimental environment for this study was based on PyTorch 1.11.0 and Python 3.8, running on the Ubuntu 20.04 operating system. In terms of hardware configuration, the experiment utilized an RTX 4090 GPU with 24GB of video memory and a 16-core Intel(R) Xeon(R) Platinum 8481C CPU. During the experiment, the input image size was uniformly set to 224×224, the number of training epochs was 40, the initial learning rate was set to 5e-5, the weight decay coefficient was 0.005, and the AdamW optimizer was selected. A learning rate scheduler was introduced to dynamically adjust the learning rate, thereby improving the model's convergence speed and overall performance. To further optimize the model's performance, multiple sets of comparative experiments were conducted for key hyperparameters such as the learning rate and batch size. The impact of different parameter configurations on the model's performance was systematically evaluated, and the optimal hyperparameter combination was ultimately determined. Through this process, the model was ensured to achieve the best balance between performance and efficiency in plant disease recognition tasks.

C. Evaluation Metrics

In this experiment, in order to comprehensively evaluate the performance of the Swin-Axial Transformer model in the plant disease detection task, the following key evaluation metrics were used: Precision, Recall, F1, model parameters, and GMAC. The comprehensive analysis of these metrics helps to thoroughly assess the performance of the Swin-Axial Transformer model in plant disease detection.

D. Experimental Results

The Swin-Axial Transformer model proposed in this paper underwent comprehensive experimental validation on the PlantDoc dataset, and its classification performance and convergence behavior were thoroughly evaluated. The specific results are analyzed as follows.

The overall classification performance of the Swin-Axial Transformer model, as shown in Table 1, yielded an average precision of 79.52%, recall of 76.69%, and F1 score of 77.78% on the PlantDoc dataset. This result indicates that the Swin-Axial Transformer model can provide high classification accuracy in multi-class plant disease detection

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tasks while ensuring good model robustness and recall ability. The classification performance for different categories is presented in Table 1. Among them, categories 16 and 23 achieved a precision, recall, and F1 score of 100%, clearly demonstrating that the model's classification ability for these categories reached an optimal level. This is likely due to the high significance of the features in these categories, which the model could capture more effectively. In contrast, category 7 has the lowest F1 score at 25.00%, which could be influenced by an imbalanced data distribution or the complexity of the features. To mitigate the impact of data imbalance, future research could consider using data augmentation techniques such as rotation, flipping, cropping, and color jitter to increase the sample size of minority categories or apply class balancing methods to enhance the model's ability to recognize these categories. Moreover, the F1 scores of most categories are concentrated in the range of 70%-95%, indicating that the model exhibits good adaptability for fine-grained classification tasks.

The specific distribution of classification performance is shown in Figure 8. The confusion matrix visually reflects the classification results of the model on the validation set. The prediction results for most categories are concentrated along the main diagonal, indicating that the model can accurately classify these categories. However, a few categories (such as category 7 and category 11) exhibit a higher misclassification frequency in the non-diagonal positions of the confusion matrix, indicating some shortcomings in feature extraction and discrimination capabilities for these categories. To address this issue, future research could consider using data augmentation techniques (such as rotation, flipping, cropping, color jitter, etc.) or class balancing methods to improve the model's ability to recognize these categories.

The trends in classification precision and loss function on the validation set are shown in Figures 9 and 10, respectively. Figure 9 illustrates the change in validation precision during the training process, which rises rapidly during the first 10 training epochs and gradually stabilizes after the 10th epoch, reaching its peak at the 37th epoch. Figure 10 displays the change in loss on the validation set, with the loss value decreasing quickly in the early stages and then remaining steady in the later stages. This trend indicates that the Swin-Axial Transformer model has good convergence and generalization ability, achieving stable and excellent performance in a short training time.

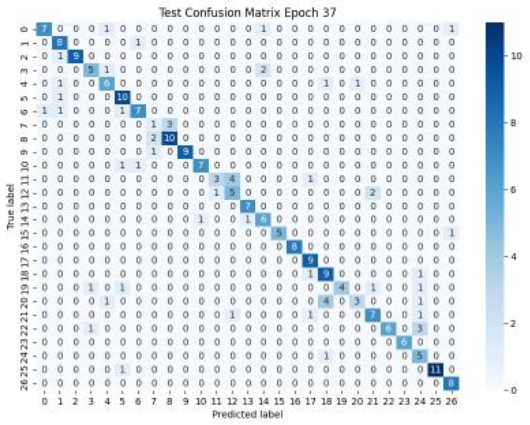


FIGURE 8. Confusion matrix

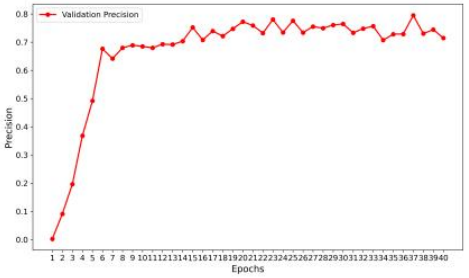


FIGURE 9. Validation precision curve

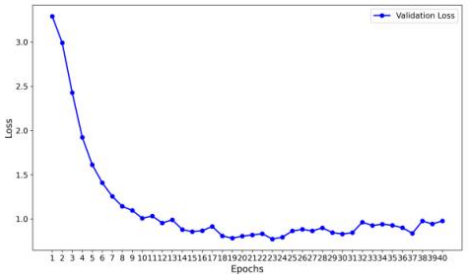


FIGURE 10. Validation loss curve

TABLE I
PLANTDOC CLASSIFICATION PERFORMANCE OF THE DATASET

Class	Precision	Recall	F1
0	87.50	70.00	77.78
1	66.67	88.89	76.19
2	100	90.00	94.74
3	71.43	62.50	66.67
4	66.67	66.67	66.67
5	71.43	90.91	80.00
6	77.78	70.00	73.69
7	25.00	25.00	25.00
8	76.92	83.33	80.00
9	100	90.00	94.74
10	87.50	77.78	82.35
11	75.00	37.50	50.00
12	50.00	62.50	55.56
13	87.50	100	93.33
14	66.67	75.00	70.59
15	100	83.33	90.91
16	100	100	100
17	75.00	100	85.71
18	60.00	81.82	69.23
19	100	50.00	66.67
20	75.00	100	85.71

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Class	Precision	Recall	F1
21	70.00	70.00	70.00
22	100	60.00	75.00
23	100	100	100
24	43.67	83.33	57.31
25	100	91.67	95.65
26	80.00	100	88.89
ALL	79.52	76.69	78.08

E. Comparative Experiments

To comprehensively validate the performance of the propose d Swin-Axial Transformer model in plant disease detection ta sks, detailed comparative experiments were conducted agains t state-of-the-art deep learning models. The comparison inclu ded classic network architectures such as EfficientNet [22], ShuffleNet [23], ResNet [24], GoogLeNet [25], MobileNet [26], DenseNet [27], MobileViT [28], ConvNeXt[29], Swin Transformer[30], and models specifically designed for plant disease detection tasks, such as T-cnn(ResNet-101) [31] and ICVT [32].

As shown in Table 2, the Swin-Axial Transformer outperformed its competitors on key classification performance metrics: Precision, Recall, and F1 Score. The F1 Score of the Swin-Axial Transformer reached 78.08%, exceeding that of DenseNet-201 (75.13%) and Swin Transformer Ting (75.00%). Additionally, compared to models with similar performance, the Swin-Axial Transformer demonstrated significant efficiency advantages in terms of computational resource usage, with a parameter count of 14.87 M and computational complexity of 2.96 GMac, which are substantially lower than those of DenseNet-201 (18.14 M, 4.39 GMac) and Swin Transformer Ting (27.54M, 4.38 GMac). This indicates that the Swin-Axial Transformer achieves a superior balance between performance and efficiency.

Compared to other Transformer-based deep learning models, the Swin-Axial Transformer achieved further optimization in both classification performance and computational efficiency. Its F1 Score improved by 2.45% over Swin Transformer Ting, while reducing the parameter count by 46%. This improvement is attributed to the effective design of the TokenEmbedder and AxialFormer Block modules, which significantly enhance the model's ability to model multi-scale features. In contrast, the Transformer base model performed poorly, with an F1 Score of only 55.58%, a parameter count of 85.82 M, and computational complexity of 16.88 GMac, demonstrating its limited suitability for plant disease detection tasks.

When compared to the DenseNet series, although DenseNet-201 achieved an F1 Score of 75.13%, close to that of the Swin-Axial Transformer, it required 18.14 M parameters and 4.39 GMac computational complexity, both of which are higher than those of the Swin-Axial Transformer. Similarly, DenseNet-161 achieved an F1 Score of 74.98%, but its parameter count (26.53 M) and computational complexity (7.85 GMac) were significantly higher than those of the Swin-Axial Transformer.

For lightweight models, MobileNetV3 Small and the ShuffleNet series have smaller parameter counts and computational complexity. For example, ShuffleNet v1.0 has only 1.28 M parameters, but its F1 Score was just 58.52%, which is insufficient for high-accuracy plant disease detection tasks. Similarly, MobileNetV3 Small achieved an F1 Score of 52.48%, significantly lagging behind the Swin-Axial Transformer. While these lightweight models offer advantages in resource-constrained scenarios, their classification performance is far below that of the Swin-Axial Transformer. By contrast, the Swin-Axial Transformer delivers significant performance improvements under low computational complexity conditions, demonstrating its strong practicality.

The T-CNN(ResNet-101) and ICVT models, specifically designed for plant disease detection tasks, perform well in classification, with T-CNN achieving a precision of 74.44% and ICVT(ResNet-101) achieving 77.23%. In comparison, the Swin-Axial Transformer outperforms both T-CNN and ICVT(ResNet-101) in precision, further validating its superior performance in plant disease detection tasks.

As shown in Table 2, the Swin-Axial Transformer demonstrates excellent classification performance and significant computational efficiency advantages in plant disease detection tasks. Compared to high-performance models such as DenseNet and Swin Transformer, the Swin-Axial Transformer not only improved classification performance but also significantly reduced parameter count and computational complexity. Furthermore, compared to lightweight models and task-specific plant disease detection models, the Swin-Axial Transformer achieved a better balance between performance and efficiency. This makes it particularly suitable for resource-constrained real-world scenarios, providing an efficient and reliable solution for large-scale plant disease detection tasks.

TABLE II
PCOMPARES THE EXPERIMENTAL RESULTS

Model	Precision	Recall	F1	parameters（M）	Glops（GMac）
EfficientNet-B0	69.78	69.92	69.85	4.04	0.406
EfficientNet -B1	73.09	70.76	71.90	6.55	0.599
EfficientNet-B2	73.24	69.92	71.54	7.72	0.689
ShufflenetV2 ×1.0	70.21	66.91	58.52	1.28	0.151
ShufflenetV2 ×2.0	72.82	69.15	70.93	5.4	0.594
ResNet-34	70.08	66.52	68.25	21.3	3.68
ResNet-50	72.58	68.64	70.55	23.56	4.13
GoogLeNet	70.10	68.22	69.15	10.01	1.51
MobileNetV3-Small	55.23	50.00	52.48	1.55	59.63
MobileNetV3-Large	62.04	58.05	59.98	4.24	229.47
DenseNet-121	74.64	72.03	73.31	6.98	2.9
DenseNet-201	77.05	73.31	75.13	18.14	4.39
DenseNet-161	76.28	73.73	74.98	26.53	7.85
MobileViT-S	73.22	71.62	72.41	4.95	1.56

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Model	Precision	Recall	F1	parameters（M）	Glops（GMac）
MobileViT-XS	74.17	69.92	71.98	1.94	0.80
ConvNeXt-Tiny	73.15	73.74	73.44	27.84	4.49
ConvNeXt-Small	71.54	70.71	71.12	49.48	8.73
Swin Transformer-T	76.80	73.30	75.00	27.54	4.38
Transformer-base	54.43	56.77	55.58	85.82	16.88
T-cnn(ResNet-101)	74.44	-	-	-	-
ICVT	77.23	-	-	-	-
Swin-Axial Transformer	79.52	76.69	78.08	14.87	2.96

F. Ablation Experiments

To evaluate the impact of different modules in the Swin-Axial Transformer model on the overall classification performance, this study designed eight sets of comparative experiments. By systematically removing or replacing key modules, the contribution of each module to the model's performance was carefully analyzed. The experimental results detailed the impact of these modifications on model parameters, computational complexity, and F1 score (see Table 3).

The first experiment used Swin-T as the baseline model, with an F1 score of 75.02%, a parameter count of 27.54 M, and a computational complexity of 4.38GMac. Although the baseline model demonstrated strong global feature extraction ability, its parameter count and computational complexity were relatively high.

The second experiment further verified the importance of the TokenEmbedder module by removing it. The results showed that the F1 score dropped to 71.68%, highlighting the crucial role of the TokenEmbedder module in enhancing model performance. Although the model's parameter count decreased to 15.14 M and computational complexity dropped to 3.16 GMac, the absence of this module significantly affected the classification performance.

Experiments 3 to 6 deeply verified the unique advantages of the module in this paper in plant disease feature modeling by replacing the TokenEmbedder module with mainstream attention mechanisms such as CBAM [33], EMA [34], ECA[35], and CoTAttention [36]. The comparison results showed that existing attention mechanisms have significant limitations in balancing multi-scale feature fusion efficiency and computational complexity. The CBAM module (Experiment 3), as a combination of channel and spatial attention, although it enhanced feature selection ability through a dual attention mechanism, its F1 score (74.32%) was still 3.76% lower than the model in this paper. This is because CBAM relies on independent attention calculations for each channel and each spatial location. When dealing with multi-scale disease spots commonly found in plant disease images (such as the coexistence of millimeter-level spots and centimeter-level spots), it is difficult to achieve efficient cross-scale feature interaction, and its computational complexity (3.16GMac) is not significantly lower than that of the TokenEmbedder module. The performance decline of the EMA and ECA modules (Experiments 4 and 5) was more obvious, with F1 scores of 72.12% and 71.87% respectively, exposing the limitations of single-dimensional attention mechanisms: EMA only focuses on weight adjustment between channels, ignoring the key role of spatial features such as disease spot morphology and texture; although ECA optimizes computational efficiency through dimensionality reduction, it is limited by a fixed receptive field and cannot capture the structural associations of disease regions at different scales, resulting in insufficient feature discrimination under complex backgrounds. The CoTAttention module (Experiment 6) learns a dynamic attention matrix by mining contextual information around the key, effectively strengthening the model's representation ability, achieving an F1 score of 72.64%, which outperforms the EMA and ECA modules. However, this module has obvious disadvantages in computational resource consumption. Both the parameter count (16.40 M) and computational complexity (3.42 GMac) are higher than those of the model in this paper, failing to achieve a good balance between feature expression and computational efficiency.

In contrast, the TokenEmbedder module, through a multi-scale depth convolution kernel group and a fixed-number token generation mechanism, while improving feature extraction efficiency, maintains low computational complexity. This module can efficiently fuse multi-scale features when processing plant disease images and avoids the deficiencies of a single attention mechanism in cross-scale feature interaction and computational resource consumption, thus achieving a good balance between F1 score and computational efficiency.

The seventh experiment replaced the Axial Transformer module with the standard Transformer module, providing a direct performance comparison between the Axial-Squeeze Attention mechanism and the traditional multi-head attention mechanism. The experimental results showed that although the standard Transformer module, with a global self-attention mechanism, achieved an F1 score of 76.53%, which is lower than the proposed model, it had a significant disadvantage in computational efficiency. Its parameter count was 15.58 M, and the computational complexity was 3.01 GMac, both of which were higher than the proposed model's 14.87 M and 2.96 GMac, respectively.

Unlike the standard Transformer's global self-attention mechanism (with a computational complexity of $O(N^2)$, the Axial Transformer module proposed in this paper uses the Axial-Squeeze Attention mechanism, which limits self-attention calculations to local regions along the horizontal and vertical axes. By compressing features along each axis and fusing information across axes, the computational complexity is optimized to $O(N)$. Moreover, the Axial-Squeeze Attention combines the Axial Compression module and the Detail Enhancement module, which efficiently extracts global semantic features and supplements local detail

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information. The synergy of these two modules significantly enhances the model's feature modeling capabilities in complex backgrounds and multi-scale scenarios.

TABLE III
RESULTS OF THE ABLATION EXPERIMENTS

Experiments	Precision	Recall	F1	Parameters（M）	Glops（GMac）
1	76.80	73.30	75.02	27.54	4.38
2	74.01	69.49	71.68	15.14	3.16
3	75.36	73.30	74.32	15.11	3.16
4	74.47	69.91	72.12	15.11	3.17
5	73.81	70.03	71.87	15.11	3.16
6	74.15	71.18	72.64	16.40	3.42
7	77.80	75.30	76.53	15.58	3.01
8	79.52	76.69	78.08	14.87	2.96

G. The Impact of Regularization Methods on Model Performance

In deep learning models, regularization methods enhance the model's generalization ability by mitigating overfitting. To investigate the impact of regularization on the performance of the Swin-Axial Transformer model in plant disease detection tasks, this study compared the model's performance under L1 regularization, L2 regularization, and without regularization. The detailed experimental results are shown in Table 4.

When using L1 regularization, the model achieved an F1 Score of 78.08%, with Precision of 79.52% and Recall of 76.69%, representing the best performance among the tested conditions. The results indicate that L1 regularization effectively suppresses overfitting by sparsifying parameters, thereby improving model performance.With L2 regularization, the model achieved an F1 Score of 76.29%, with Precision of 78.55% and Recall of 74.16%, slightly lower than L1 regularization. This suggests that while L2 regularization effectively penalizes weights to enhance generalization ability, its impact is somewhat less pronounced than L1 regularization in this model.In the absence of regularization, the model achieved an F1 Score of 75.96%, with Precision of 78.53% and Recall of 75.45%. Compared to L1 regularization, the F1 Score decreased by 2.12%, demonstrating the critical role of regularization methods in improving model performance and preventing overfitting.

TABLE IV
EFFECT OF THE REGULARIZATION METHOD ON THE MODEL PERFORMANCE

Regularization Methods	Precision	Recall	F1
L1	79.52	76.69	78.08
L2	78.55	74.16	76.29
No	78.53	75.45	75.96

H. Exploration of the Impact of Token Dimensions and Quantities on Model Performance

In this study, we systematically explored the effects of different Token Dimensions and Token Quantities in the TokenEmbedder module on model classification performance, parameter count, and computational cost (GFLOPs). By adjusting the Token Dimensions (128, 256, 512, and 768) and Token Quantities (36, 64, 100, and 225), we analyzed the changes in Precision, Recall, F1 Score, model parameters, and GFLOPs. The experimental results demonstrate that both Token Dimensions and Quantities significantly influence model performance, especially with clear trade-offs between classification accuracy, computational complexity, and resource consumption.

With Token Dimensions increasing from 128 to 768, the F1 Score and resource consumption showed significant changes. When the Token Dimension was 256, the model achieved its best performance, with an F1 Score of 78.08%, Precision of 79.52%, and Recall of 76.69%. Further increases in dimensions to 512 and 768 caused F1 Scores to drop to 76.14% and 73.09%, respectively, suggesting that excessively large dimensions lead to feature redundancy, thereby affecting classification performance. Increasing Token Dimensions also significantly raised parameter count and GFLOPs. For example, as the Token Dimension increased from 128 to 768, the parameter count rose from 13.23 M to 27.97 M, and GFLOPs increased from 2.74 GMac to 4.31 GMac. However, performance gains were limited. A Token Dimension of 256 achieved the optimal F1 Score while maintaining relatively low computational complexity (2.93 GMac), striking the best balance between performance and resource consumption.

With Token Dimensions fixed at 256, increasing the Token Quantity from 36 to 225 improved the F1 Score from 75.73% to 76.44%, with simultaneous increases in Precision and Recall. A greater number of tokens helped capture fine-grained features, thereby improving classification performance. However, increasing the Token Quantity also raised GFLOPs. For instance, as Token Quantity increased from 36 to 225, GFLOPs rose from 2.89 GMac to 3.17 GMac. The balance between performance and resource consumption depends on the specific scenario. At a Token Quantity of 64, the F1 Score reached 78.08%, and GFLOPs were 2.93 GMac, achieving a favorable trade-off between performance and resource efficiency.

TABLE V
EFFECT OF TOKEN SIZE AND COUNT ON MODEL PERFORMANCE AND COMPUTATIONAL EFFICIENCY

Dimension	Quantity	Precision	Recall	F1	Parameters（M）	Glops（GMac）
64	128	76.02	73.72	74.85	13.23	2.74
64	256	79.52	76.69	78.08	14.87	2.93
64	512	78.23	74.15	76.14	20.11	3.5
64	768	75.10	71.19	73.09	27.97	4.31
36	256	77.37	74.15	75.73	14.86	2.89
100	256	77.56	74.15	75.82	14.87	2.98

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225	256	77.93	75.00	76.44	14.90	3.17

I. Impact of different depthwise convolution kernels in the TokenEmbedder module on model performance

To explore the impact of deep convolution kernel sizes on the performance of the Swin-Axial Transformer model, this study designed comparative experiments using the control variable method. The aim was to analyze the feature extraction capabilities and classification performance of different convolution kernel combinations (dual-scale and triple-scale) in plant disease detection tasks.

As shown in Table 6, the experimental results indicate that when using 5 × 5, 7 × 7, and 9 × 9 triple-scale deep convolution kernels, the model achieved a precision of 79.52%, recall of 76.69%, and an F1 score of 78.08% on the PlantDoc dataset, showing a significant improvement over other kernel combinations. Compared to combinations with larger kernels (such as 11 × 11) or smaller kernels (such as 3 × 3), the medium-sized kernel configuration (5 × 5, 7 × 7, 9 × 9) strikes an optimal balance between computational efficiency and feature representation. This configuration avoids the fragmentation of local features caused by insufficient receptive fields with smaller kernels and the risk of increased background noise and computational complexity with larger kernels. The experimental conclusion validates the crucial role of multi-scale deep convolution kernels in the TokenEmbedder module, indicating that the fusion of features with differentiated receptive fields effectively enhances disease detection accuracy in complex scenarios.

TABLE VI
COMPARISON OF MODEL PERFORMANCE UNDER DIFFERENT DEPTHWISE CONVOLUTION KERNEL CONFIGURATIONS

Convolution kernel configuration	Precision	Recall	F1
3 × 3, 5 × 5	76.10	73.72	74.89
3 × 3, 7 × 7	77.18	73.30	75.19
3 × 3, 5×5	76.09	74.23	75.15
3 × 3, 5 × 5, 7 × 7	78.24	75.02	76.60
7 × 7, 9 × 9, 11 × 11	77.41	73.72	75.52
5 × 5, 7 × 7, 9 × 9	79.52	76.69	78.08

J. Generalization Ability and Real-Time Performance Testing

This paper evaluates the generalization ability and stability of the Swin-Axial Transformer model using both the PlantVillage dataset and a custom fusion dataset, comparing its performance with that of other existing models. In the experiment conducted on the PlantVillage dataset, as shown in Table 7, the Swin-Axial Transformer performed exceptionally well, achieving an precision of 99.79%, a recall rate of 99.78%, and an F1 score of 99.78, clearly outperforming other comparative models. These results demonstrate that the Swin-Axial Transformer not only effectively balances classification precision and recall but also offers significant advantages in plant disease detection tasks, providing strong evidence of its powerful application potential in solving standardized problems.

In the experiment conducted on the custom fusion dataset, as shown in Table 8, the Swin-Axial Transformer demonstrated outstanding classification performance, achieving a precision of 95.69%, a recall rate of 95.38%, and an F1 score of 95.53%. Additionally, it exhibited strong real-time processing capabilities, achieving a frame rate (FPS) of 176. Compared to other models such as Transformer-base (FPS: 78) and Swin Transformer Ting (FPS: 136), the Swin-Axial Transformer showed a clear advantage in computational efficiency, excelling in applications that require a balance between high accuracy and low latency. Although ResNet-50 achieved a higher FPS of 214, the Swin-Axial Transformer delivered near real-time performance while maintaining high classification accuracy and low computational complexity. In contrast, the Swin-Axial Transformer offers a significant advantage in terms of model parameter count and computational complexity, making it particularly well-suited for environments with limited computational resources. Therefore, the Swin-Axial Transformer provides an efficient solution for real-time plant disease detection tasks, achieving an excellent balance between accuracy and efficiency.

TABLE VII
PLANTVILLAGE DATASET EXPERIMENTAL RESULTS

Model	Precision	Recall	F1
CNN [37]	98.27	98.01	98.14
Attention and theresidual network [38]	96.20	95.60	95.90
DenseNet along with attention approach [39]	89.59	86.71	88.13
PlantViT [40]	98.24	98.33	98.28
ConvViT-S [41]	97.80	98.54	98.17
TLMViT	98.72	98.76	98.74
Swin-Axial Transformer	99.79	99.78	99.78

TABLE VIII
SELF-CONSTRUCTED FUSION DATASET EXPERIMENTAL RESULTS

Model	Precision	Recall	F1	FPS
Transformer-base	85.02	84.93	84.97	78
Swin Transformer-T	93.34	93.28	93.31	136
Resnet-50	92.86	92.64	92.75	214
Swin-Axial Transformer	95.69	95.38	95.53	176

IV. Conclusion

The Swin-Axial Transformer model proposed in this paper effectively addresses the high computational costs, complex backgrounds, and multi-scale feature extraction challenges in plant disease detection tasks. Unlike traditional object or pattern detection tasks, plant disease detection not only requires identifying and classifying disease types but also

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must deal with complex natural environmental backgrounds, diverse disease characteristics, and differences across plants' various growth stages and species. Plant leaves, fruits, or stems may exhibit different disease symptoms under varying environmental conditions, and these symptoms often intertwine with complex backgrounds, further complicating disease detection.To effectively tackle this challenge, this paper introduces the TokenEmbedder module, which reduces the number of tokens and combines multi-scale deep convolution to efficiently extract image features. This improvement significantly lowers computational complexity while optimizing the feature extraction process. The module provides a more compact and efficient input for subsequent models, further enhancing overall computational efficiency.
Furthermore, the Axial Transformer module introduced in this paper reduces computational complexity by restricting self-attention calculations to local regions along the axial directions. At the same time, by combining the Axial Compression module and the Detail Enhancement module, the Axial Transformer efficiently extracts global semantic features and supplements local detail information. The synergistic effect of these two modules significantly enhances the feature modeling capabilities of the Axial Transformer in complex backgrounds and multi-scale scenarios.

Experimental results show that the Swin-Axial Transformer outperforms multiple datasets with significant advantages. For instance, on the PlantDoc dataset, the model achieved a precision of 79.52%, which is 5.02% higher than the T-CNN (ResNet-101) model at 74.50%; on the PlantVillage dataset, the precision was 99.79%, which is an improvement of 1.91% over the ConvViT-S model; on the self-constructed fusion dataset, the precision was 95.69%, which is 3.45% higher than the Swin Transformer-T model. Additionally, the model has 14.87 M parameters and a computational load of 2.96 GMac, which represents a 46% reduction in parameters and a 32% reduction in computational load compared to the Swin Transformer-T model, significantly improving computational efficiency. These results indicate that the Swin-Axial Transformer not only surpasses several mainstream models in classification accuracy but also demonstrates significant advantages in terms of computational resource consumption. The model presented in this paper provides an efficient and reliable solution for large-scale crop disease detection, with broad application prospects.

However, this study also acknowledges certain limitations. First, although the Swin-Axial Transformer has been optimized for computational efficiency, it still requires substantial computational resources when processing high-resolution images and large-scale datasets. Second, the current model architecture is primarily optimized for specific datasets, and there may be a risk of performance degradation when extended to more diverse diseases and crops. Furthermore, although the model performs excellently on several experimental datasets, its robustness may be affected in more complex backgrounds, lighting changes, and other scenarios, leading to unstable performance in real-world applications. Finally, although the Swin-Axial Transformer has made progress in real-time processing, its real-time response capability may still be constrained by computational resources in resource-limited environments, especially in real-time disease monitoring systems, where it may not fully meet the demands for low latency and efficient inference.

Future research can improve the Swin-Axial Transformer model's generalization and scalability in several directions. First, to further optimize computational efficiency and real-time performance, lightweight model techniques such as pruning, quantization, and model distillation can be employed to reduce computational resource consumption, while combining hardware acceleration (such as GPUs or TPUs) to improve inference speed. These optimizations will make the model suitable not only for plant disease detection tasks but also extend its applicability to real-time inference fields, such as video surveillance, autonomous driving, and more. Additionally, adopting distributed computing architectures and edge computing strategies can distribute computational tasks across multiple nodes, reducing the burden on a single device and ensuring low-latency and efficient inference in resource-constrained environments, further enhancing the model's versatility.Second, to enhance the model's scalability and generalizability, multi-task learning and cross-domain transfer learning strategies can be implemented, allowing the Swin-Axial Transformer to maintain high performance and stability when handling a broader range of object and pattern detection tasks. For example, pretraining the model on different object detection datasets and fine-tuning it for new tasks will help the model handle complex backgrounds and multi-scale issues in plant disease detection, as well as perform excellently in fields like medical imaging, industrial defect detection, and others. Through this approach, the model can more flexibly adapt to variations across different domains.Finally, to improve the model's robustness, future research can focus on optimizing data augmentation techniques and self-supervised learning methods, ensuring that the model maintains efficient detection capabilities even in complex backgrounds, lighting changes, and partial occlusions. These improvements will enhance the Swin-Axial Transformer's adaptability and generalization across various complex environments, meeting the needs of a wider range of object and pattern detection tasks, and providing more stable and precise solutions for large-scale, real-time monitoring systems.

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